



Mathematics of the Simple Camera

Overview

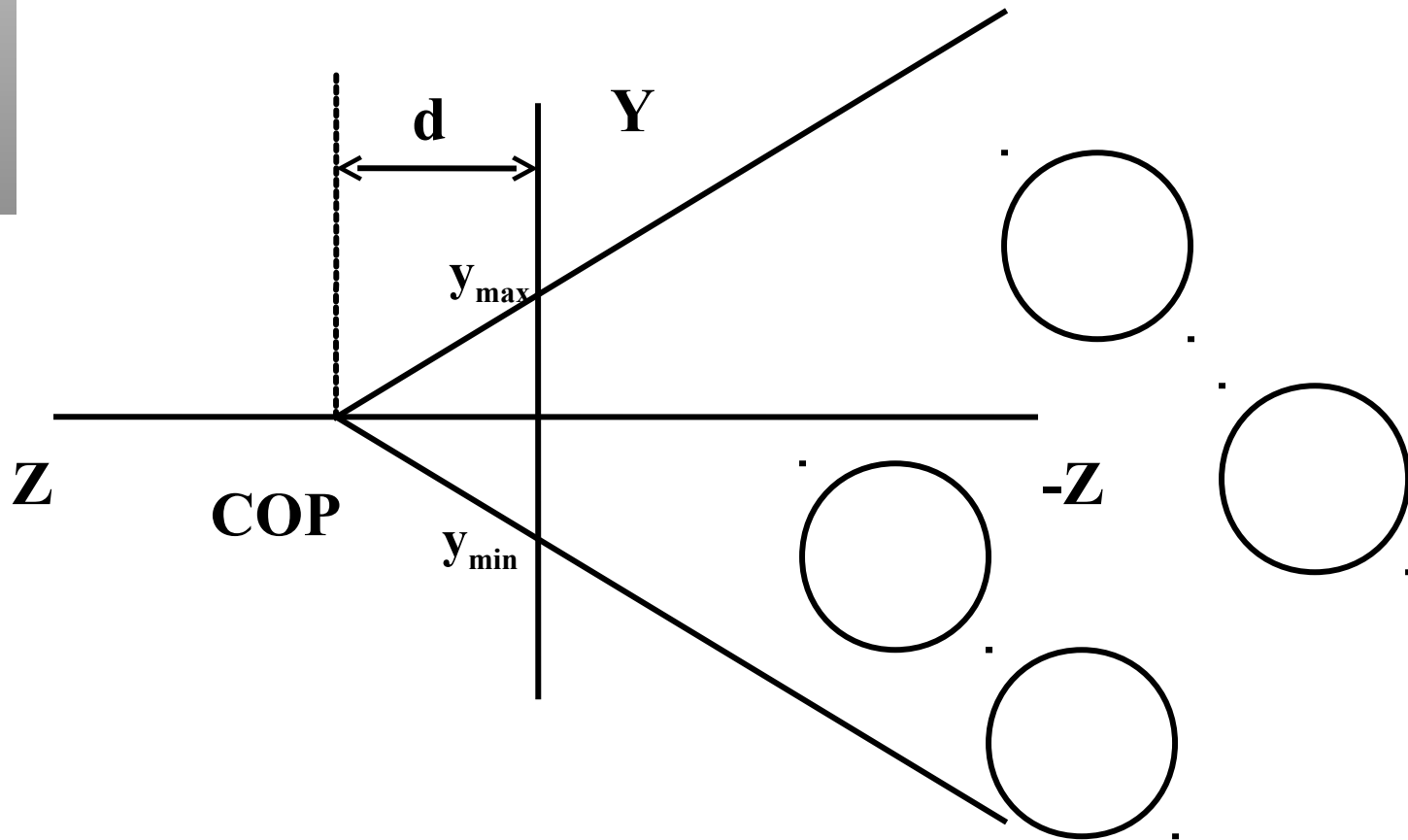
■ *Simple Camera*

- *Scenes with spheres*
- *COP on +z*

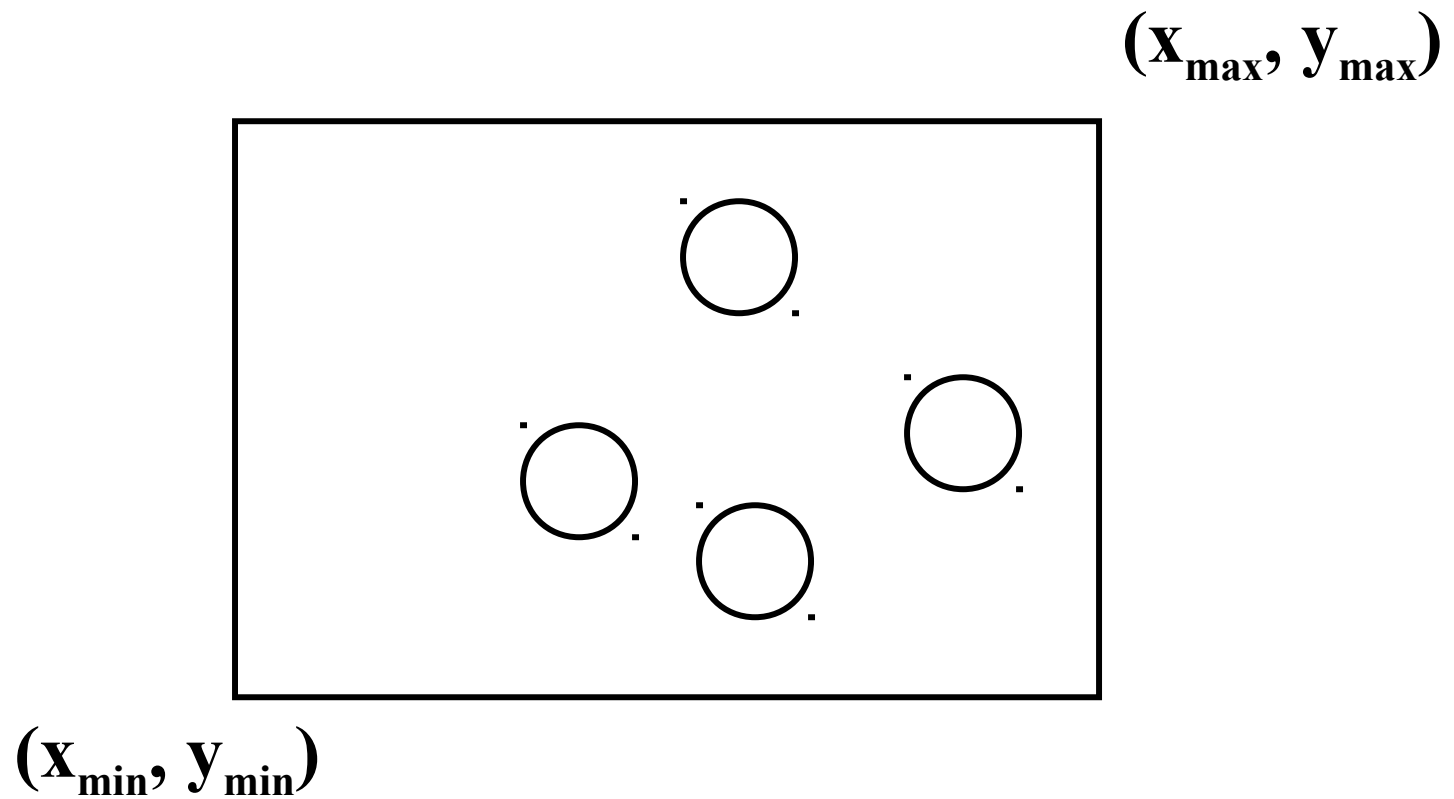
COP = Centre Of Projection

Simple Camera (Cross Section)

COP = Centre Of Projection

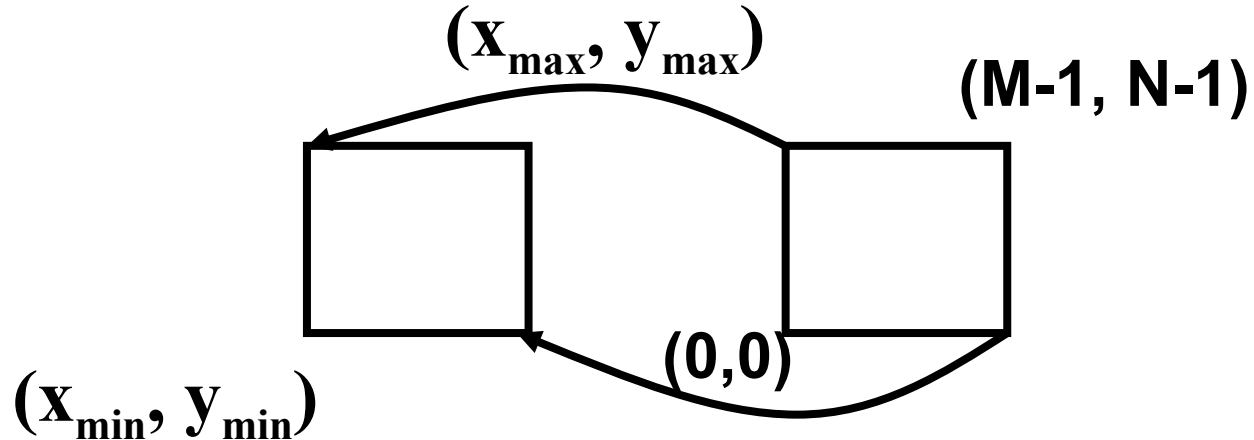


View From the Camera



Forming the Rays

- *Map screen pixels (M by N window) to points in camera view plane*



Forming the Rays

- *Consider pixel i,j*
- *It corresponds to a rectangle*
$$\text{width} = (x_{\max} - x_{\min}) / M$$
$$\text{height} = (y_{\max} - y_{\min}) / N$$
- *Our ray goes through the centre of the pixel*
- *Thus the ray goes through the point*
$$(x_{\min} + \text{width} * (i + 0.5), y_{\min} + \text{height} * (j + 0.5), 0.0)$$

Forming the Rays

- *Thus the ray from the COP through pixel i,j is defined by*

$$\begin{aligned} p(t) = (x(t), y(t), z(t)) = \\ (t*(xmin + width*(i+0.5)), \\ t*(ymin + height*(j+0.5)), \\ d - t*d) \end{aligned}$$

Ray Casting

- *Intersection of Sphere and line (sphere at origin)*
- *Substitute the ray equation in the sphere equation and solve!*
- *Get an equation in t of the form*

$$At^2 + 2Bt + C = 0$$

Ray Casting

If $B^2 - AC < 0$ then the ray doesn't intersect the sphere.

If $B^2 - AC = 0$ the ray is tangent to the sphere

If $B^2 - AC > 0$ then there are two roots given by

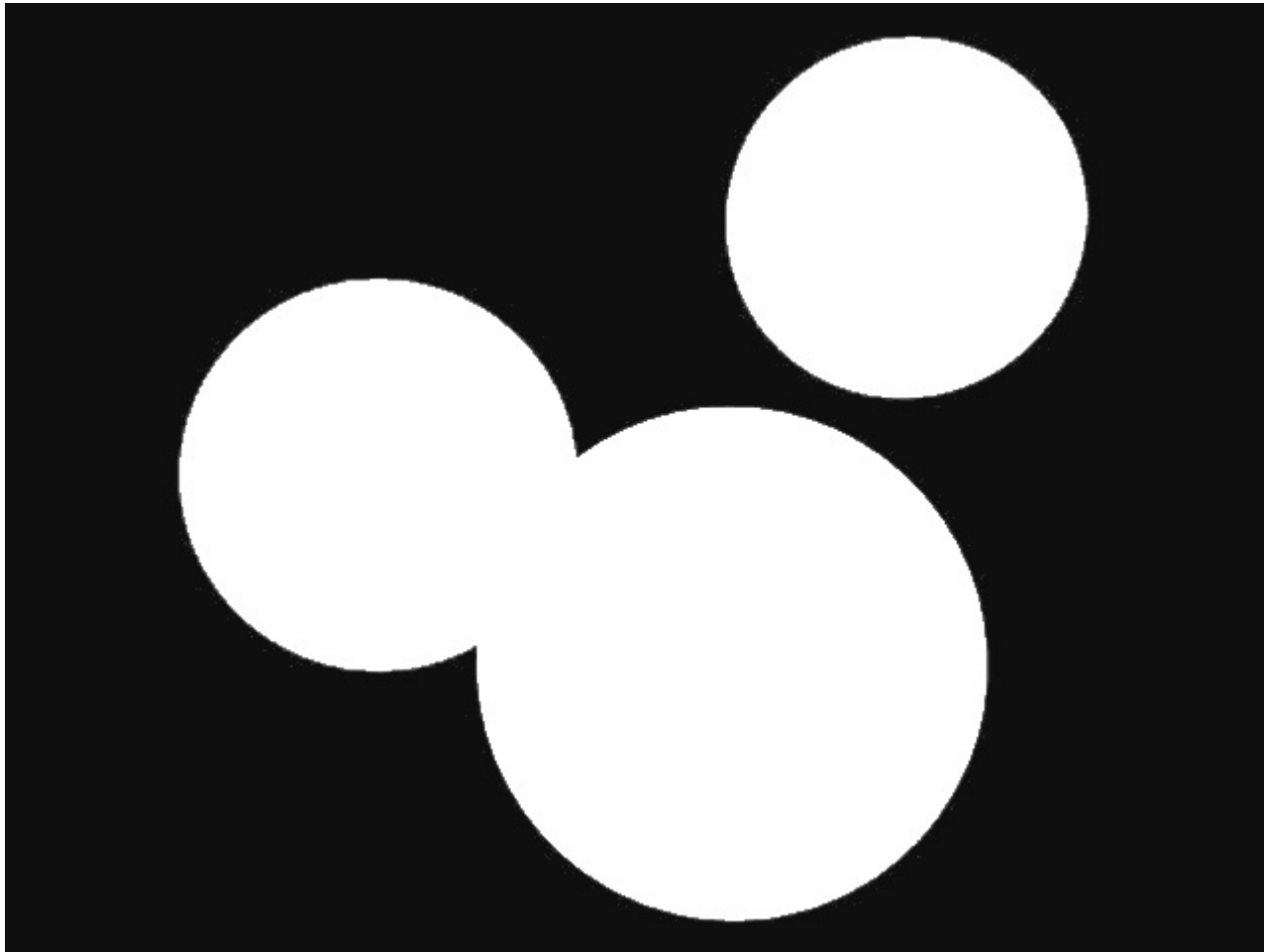
$$t = (-B \pm \sqrt{B^2 - AC})/A$$

choose the lowest value one (the one closest to the COP)

Ray Casting

- ***Intersection of Sphere and line (general case)***
 - ***Sphere is centred at (a,b,c)***
 - ***Translate the start of the ray by $(-a,-b,-c)$***
 - ***Proceed as before***

The Image - Detection



Conclusions

- ***We can now draw images***
 - ***Forming rays from the camera***
 - ***Intersecting those rays with objects (spheres) in the scene***
- ***But***
 - ***No colour – merely binary detection operation***
 - ***Camera is static - at the moment we must move the objects in front of the camera to be able to see them***
 - ***Need more interesting scenes!***